```
/* ===== Student Attendance Tracker ===== */
const STATUSES = ["Present", "Absent", "Late", "On Duty"];
const ATTENDED = ["Present", "Late", "On Duty"]; // statuses counted as attended

function load(key, fallback) {
  try { const v = JSON.parse(localStorage.getItem(key)); return v === null ? fallback
: v; } catch (e) { return fallback; }
}
let students = load("sat_students", null);
let attendance = load("sat_attendance", {}); // { "2026-10-27": { "BAI101": "Present"
} } -> one entry per student per date
let threshold = Number(load("sat_threshold", 75)) || 75;

/* Sample data so the app is not empty on first open (delete or edit freely) */
if (!students) {
  students = [
    { id: "BAI101", name: "Aarthi Devi", dept: "BSc AI" }, { id: "BAI102", name:
"Divya Lakshmi", dept: "BSc AI" },
    { id: "BAI103", name: "Fathima Rizwana", dept: "BSc AI" }, { id: "BAI104", name:
"Harini Priya", dept: "BSc AI" },
    { id: "CSC201", name: "Janani Suresh", dept: "BSc CS" }, { id: "CSC202", name:
"Meena Kumari", dept: "BSc CS" },
    { id: "BCA301", name: "Nasreen Banu", dept: "BCA" }, { id: "BCA302", name: "Priya
Dharshini", dept: "BCA" }
  ];
  const days = ["2026-10-15","2026-10-16","2026-10-19","2026-10-20","2026-10-
21","2026-10-22","2026-10-23","2026-10-26","2026-10-27","2026-10-28"];
  const absentIdx = { BAI102: [4], BAI103: [3, 5, 6], BAI104: [2], CSC201: [0, 3, 5,
7], CSC202: [1], BCA301: [1, 2, 4, 6], BCA302: [8] };
  const lateIdx = { BAI101: [7], CSC202: [9], BCA302: [3] };
  days.forEach((d, i) => {
    attendance[d] = {};
    students.forEach((s) => {
      attendance[d][s.id] = (absentIdx[s.id] || []).includes(i) ? "Absent" :
(lateIdx[s.id] || []).includes(i) ? "Late" : "Present";
    });
  });
  threshold = 75;
  try {
    localStorage.setItem("sat_students", JSON.stringify(students));
    localStorage.setItem("sat_attendance", JSON.stringify(attendance));
    localStorage.setItem("sat_threshold", threshold);
  } catch (e) {}
}
let editingId = null;
let draft = {}; // unsaved statuses on Mark Attendance page
```

```
const $ = (id) => document.getElementById(id);
const esc = (t) => String(t).replace(/[&<>"']/g, (c) => ({ "&": "&amp;", "<": "&lt;",
">": "&gt;", '"': "&quot;", "'": "&#39;" }[c]));

function saveData() {
  try {
    localStorage.setItem("sat_students", JSON.stringify(students));
    localStorage.setItem("sat_attendance", JSON.stringify(attendance));
    localStorage.setItem("sat_threshold", threshold);
  } catch (e) {}
}

function toast(msg) {
  const t = $("toast");
  t.textContent = msg;
  t.hidden = false;
  clearTimeout(toast.timer);
  toast.timer = setTimeout(() => (t.hidden = true), 2500);
}

function formatDate(d) {
  return new Date(d + "T00:00:00").toLocaleDateString("en-GB", { day: "2-digit",
month: "short", year: "numeric" });
}

function latestDate() {
  const dates = Object.keys(attendance).filter((d) =>
Object.keys(attendance[d]).length).sort();
  return dates.length ? dates[dates.length - 1] : null;
}

function statsFor(id) {
  let total = 0, attended = 0;
  Object.keys(attendance).forEach((date) => {
    const s = attendance[date][id];
    if (s) { total++; if (ATTENDED.includes(s)) attended++; }
  });
  const percent = total ? Math.round((attended / total) * 100) : 0;
  return { total, attended, percent };
}

function renderSummary() {
  const date = latestDate();
  const day = date ? attendance[date] : {};
  const present = students.filter((s) => ATTENDED.includes(day[s.id])).length;
  const absent = students.filter((s) => day[s.id] === "Absent").length;
  let total = 0, attended = 0;
  students.forEach((s) => { const st = statsFor(s.id); total += st.total; attended +=
st.attended; });
```

```
  const overall = total ? Math.round((attended / total) * 100) : 0;
  const label = date ? "on " + formatDate(date) : "no records yet";

  $("summary").innerHTML = `
    <div class="card"><span>Total Students</span><strong>${students.length}</strong>
</div>
    <div class="card green"><span>Present Students</span><strong>${present}</strong>
<br><small>${label}</small></div>
    <div class="card red"><span>Absent Students</span><strong>${absent}</strong><br>
<small>${label}</small></div>
    <div class="card blue"><span>Overall Attendance</span><strong>${overall}%
</strong><br><small>all students, all dates</small></div>`;
}

function fillDepartments() {
  const depts = [...new Set(students.map((s) => s.dept))].sort();
  ["deptFilter", "attDept"].forEach((id) => {
    const sel = $(id), cur = sel.value;
    sel.innerHTML = '<option value="">All Departments</option>' +
      depts.map((d) => `<option value="${esc(d)}">${esc(d)}</option>`).join("");
    sel.value = depts.includes(cur) ? cur : "";
  });
  $("deptList").innerHTML = depts.map((d) => `<option value="${esc(d)}">`).join("");
}

function renderStudents() {
  const q = $("searchInput").value.trim().toLowerCase();
  const dept = $("deptFilter").value;
  const status = $("statusFilter").value;
  const date = latestDate();

  const rows = students.filter((s) => {
    if (q && !(s.name.toLowerCase().includes(q) || s.id.toLowerCase().includes(q)))
return false;
    if (dept && s.dept !== dept) return false;
    if (status === "low") return statsFor(s.id).percent < threshold &&
statsFor(s.id).total > 0;
    if (status) return date && attendance[date][s.id] === status;
    return true;
  });

  $("studentBody").innerHTML = rows.map((s) => {
    const st = statsFor(s.id);
    const low = st.total > 0 && st.percent < threshold;
    return `<tr class="${low ? "low" : ""}">
      <td>${esc(s.id)}</td><td>${esc(s.name)}</td><td>${esc(s.dept)}</td>
      <td>${st.attended} / ${st.total}</td><td>${st.percent}%</td>
      <td><span class="badge ${low ? "low" : "ok"}">${low ? "Low Attendance" :
"Good"}</span></td>
      <td>
```

```
        <button class="btn small"
onclick="showHistory('${esc(s.id)}')">History</button>
        <button class="btn small"
onclick="openStudentModal('${esc(s.id)}')">Edit</button>
        <button class="btn small danger"
onclick="deleteStudent('${esc(s.id)}')">Delete</button>
      </td></tr>`;
  }).join("");
  $("emptyMsg").hidden = rows.length > 0;
}

function openStudentModal(id) {
  editingId = id || null;
  const s = id ? students.find((x) => x.id === id) : null;
  $("studentModalTitle").textContent = s ? "Edit Student" : "Add Student";
  $("stuId").value = s ? s.id : "";
  $("stuId").disabled = !!s; // ID cannot change while editing
  $("stuName").value = s ? s.name : "";
  $("stuDept").value = s ? s.dept : "";
  $("stuError").textContent = "";
  $("studentModal").hidden = false;
}

function saveStudent() {
  const id = $("stuId").value.trim();
  const name = $("stuName").value.trim();
  const dept = $("stuDept").value.trim();
  if (!id || !name || !dept) { $("stuError").textContent = "All fields are
required."; return; }
  if (!editingId && students.some((s) => s.id.toLowerCase() === id.toLowerCase())) {
    $("stuError").textContent = "Student ID already exists."; return;
  }
  if (editingId) {
    const s = students.find((x) => x.id === editingId);
    s.name = name; s.dept = dept;
    toast("Student updated");
  } else {
    students.push({ id, name, dept });
    toast("Student added");
  }
  saveData();
  $("studentModal").hidden = true;
  refreshAll();
}

function deleteStudent(id) {
  if (!confirm("Delete this student and all attendance records?")) return;
  students = students.filter((s) => s.id !== id);
  Object.keys(attendance).forEach((d) => delete attendance[d][id]);
  saveData();
```

```
  refreshAll();
  toast("Student deleted");
}

/* ----- Mark Attendance ----- */
function attStudents() {
  const dept = $("attDept").value;
  return students.filter((s) => !dept || s.dept === dept);
}

function renderAttendance() {
  $("attBody").innerHTML = attStudents().map((s) => `
    <tr><td>${esc(s.id)}</td><td>${esc(s.name)}</td><td>${esc(s.dept)}</td>
    <td><div class="status-group">${STATUSES.map((st) =>
      `<button class="${draft[s.id] === st ? "sel-" + st.replace(" ", "") : ""}"
onclick="setStatus('${esc(s.id)}','${st}')">${st}</button>`
    ).join("")}</div></td></tr>`).join("");
}

function setStatus(id, status) { draft[id] = status; renderAttendance(); }

function markAllPresent() {
  attStudents().forEach((s) => (draft[s.id] = "Present"));
  renderAttendance();
}

function loadDraft() {
  const date = $("attDate").value;
  const existing = attendance[date];
  draft = existing ? { ...existing } : {};
  const banner = $("attBanner");
  if (existing) {
    banner.textContent = "Attendance for " + formatDate(date) + " is already
recorded. Saving will update the existing entries - no duplicates are created.";
    banner.hidden = false;
  } else {
    banner.hidden = true;
  }
  renderAttendance();
}

function saveAttendance() {
  const date = $("attDate").value;
  if (!date) { toast("Please select a date"); return; }
  if (!Object.keys(draft).length) { toast("Mark at least one student"); return; }
  const already = !!attendance[date];
  attendance[date] = { ...(attendance[date] || {}), ...draft }; // one status per
student per date
  saveData();
  refreshAll();
```

```
  loadDraft();
  toast(already ? "Existing attendance updated (no duplicates)" : "Attendance
saved");
}

/* ----- History ----- */
function showHistory(id) {
  const s = students.find((x) => x.id === id);
  const st = statsFor(id);
  $("historyTitle").textContent = s.name + " (" + s.id + ")";
  $("historyStats").innerHTML = `<span>Total days: ${st.total}</span><span>Attended:
${st.attended}</span><span>Absent: ${st.total - st.attended}</span><span>Attendance:
${st.percent}%</span>`;
  const dates = Object.keys(attendance).filter((d) => attendance[d]
[id]).sort().reverse();
  $("historyBody").innerHTML = dates.map((d) => {
    const status = attendance[d][id];
    const day = new Date(d + "T00:00:00").toLocaleDateString("en-GB", { weekday:
"long" });
    return `<tr><td>${formatDate(d)}</td><td>${day}</td><td><span class="badge
${status.replace(" ", "")}">${status}</span></td></tr>`;
  }).join("") || '<tr><td colspan="3">No records yet.</td></tr>';
  $("historyModal").hidden = false;
}

/* ----- Navigation & setup ----- */
function switchView(view) {
  document.querySelectorAll(".nav-btn").forEach((b) => b.classList.toggle("active",
b.dataset.view === view));
  $("view-students").hidden = view !== "students";
  $("view-attendance").hidden = view !== "attendance";
  if (view === "attendance") loadDraft();
}

function refreshAll() {
  fillDepartments();
  renderSummary();
  renderStudents();
  renderAttendance();
}

document.querySelectorAll(".nav-btn").forEach((b) => b.addEventListener("click", ()
=> switchView(b.dataset.view)));
$("addStudentBtn").addEventListener("click", () => openStudentModal(null));
$("saveStudent").addEventListener("click", saveStudent);
$("cancelStudent").addEventListener("click", () => ($("studentModal").hidden =
true));
$("closeHistory").addEventListener("click", () => ($("historyModal").hidden = true));
["searchInput", "deptFilter", "statusFilter"].forEach((id) => {
  $(id).addEventListener("input", renderStudents);
```

```
  $(id).addEventListener("change", renderStudents);
});
$("thresholdInput").addEventListener("input", () => {
  threshold = Number($("thresholdInput").value) || 75;
  saveData();
  renderStudents();
});
$("attDate").addEventListener("change", loadDraft);
$("attDept").addEventListener("change", renderAttendance);
$("markAllBtn").addEventListener("click", markAllPresent);
$("saveAttBtn").addEventListener("click", saveAttendance);

$("thresholdInput").value = threshold;
$("attDate").value = new Date().toISOString().slice(0, 10);
refreshAll();
```